

LESSON 05 / 第五課

Planting and Establishment: From Nursery Systems to the Flare Datum

種植與定植：從苗圃系統到幹基膨大部分的基準

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Establishment is the interval in which a transplanted tree rebuilds an absorbing root system that can pay the crown's water and carbon bills. At production scale, container, field-grown and internally girdled systems leave different root defects. At the planting unit the non-negotiable landmark is the root flare: burying it is a chronic stem and oxygen defect, not a taste. Hole geometry, backfill and correction of circling roots are the published tactics. Watson's establishment clock is root growth, not a number of calendar days. Hilbert's synthesis found mortality elevated in the first five years. A planting programme without that clock is a canopy claim without a survival budget.

定植是移植樹重建吸收根系、使其能支付樹冠水與碳帳單的區間。生產尺度上，容器、地栽與內環縊系統留下不同根系缺陷。種植單元上不可談判的地標是幹基膨大部：把它埋住是慢性莖與氧缺陷，不是品味。穴幾何、回填與矯正盤繞根是已發表策略。Watson 的定植時鐘是根系生長，不是日曆天數。Hilbert 綜述發現死亡率在首五年升高。沒有此時鐘的種植計劃，是沒有存活預算的樹冠主張。

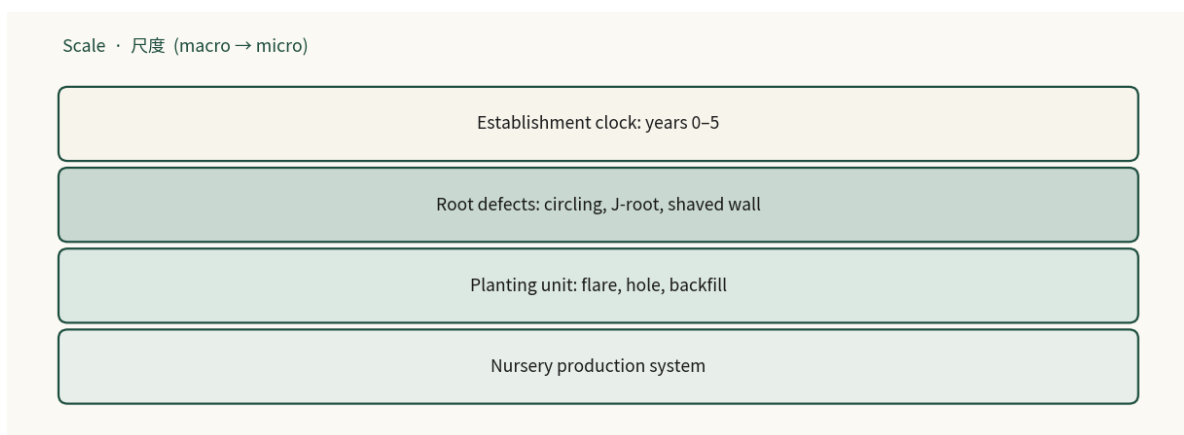


Figure 1. Nested scales from nursery production to the establishment clock.

圖1. 由苗圃生產到定植時鐘的嵌套尺度。



Figure 2. Flare datum and container-wall defects as first-order errors.

圖2. 幹基基準與容器壁缺陷作為一階錯誤。

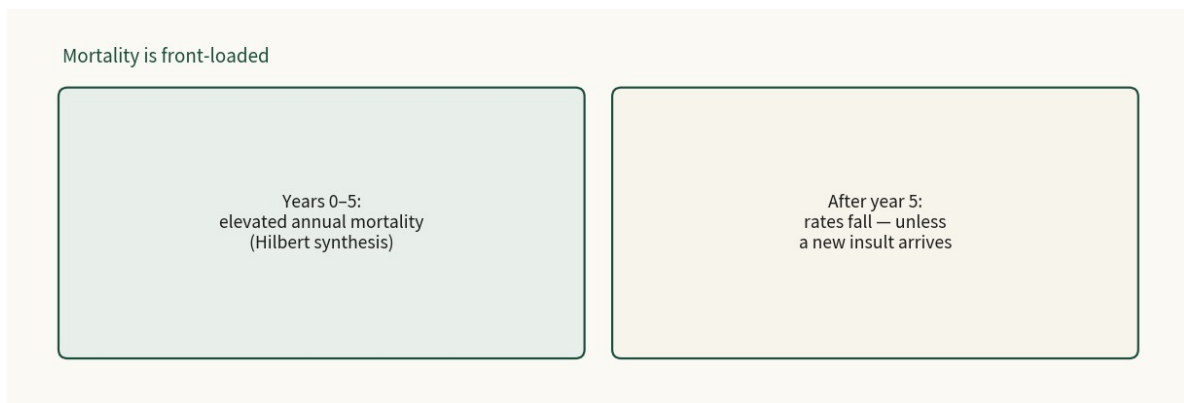


Figure 3. Mortality risk is front-loaded in years 0-5.

圖3. 死亡風險在第0-5年前置。

1. Scope

1. 範圍

The object is published transplant and establishment biology, from nursery production systems to the flare datum and the first five years.

對象是已發表的移植與定植生物學：由苗圃生產系統到幹基基準與首五年。

2. Nursery systems and the flare

2. 苗圃系統與幹基

Container walls train circling and descending roots; field-grown stock trades those defects for a severed absorbing system at harvest; air-pruning systems change the defect class rather than abolish defects [1,2]. Gilman and colleagues showed that shaving or teasing a circling container face before planting changes subsequent root architecture relative to planting the defect intact [1]. The root flare is the morphological landmark of the stem–root junction. Planting with the flare below finished grade buries bark in a hypoxic collar and invites adventitious roots above a buried structural system [2,3]. ANSI A300 Part 6 treats flare visibility at grade as a specification item [3].

容器壁訓練盤繞與下扎根；地栽苗以採收時切斷吸收系統來交換那些缺陷；氣剪系統改變缺陷類別，而不是消滅缺陷 [1,2]。Gilman 及同事顯示，種植前削去或梳開盤繞的容器面，相對原封不動種下缺陷，會改變其後根系架構 [1]。幹基膨大部是莖–根交界的形態地標。把膨大部種在完成標

高以下，等於把樹皮埋進缺氧領圈，並在被埋結構根系之上引誘不定根 [2,3]。ANSI A300 第 6 部分把標高處可見膨大部當成規格項目 [3]。

3. Hole, backfill, mulch

3. 穴、回填、覆蓋

A hole that is deep and narrow recreates a container. A hole that is wide at absorbing-root depth, with the flare set to grade, is the published geometry [3]. Amending backfill so that it becomes a pot inside native soil recreates a container hydrology. The published default is native soil, broken out of compaction, with water used to settle — not to drown. Mulch is a surface specification in A300 Part 2; volcano mulch against the stem is the opposite of that specification.

又深又窄的穴會再造一個容器。在吸收根深度上寬、並把膨大部放到標高的穴，才是已發表幾何 [3]。把回填改到變成原生土裡的一盆，會再造容器水文。已發表預設是原生土、打破壓實、用水沉實——不是淹死。覆蓋物在 A300 第 2 部分是表面規格；堆成火山抵住莖，是該規格的反面。

4. Establishment clock

4. 定植時鐘

Watson treated establishment as the rebuilding of absorbing roots into the surrounding soil, measured in root elongation and crown water status, not in a universal day-count [4]. Hilbert et al. found that annual mortality is higher in the first five years than afterwards across many urban studies [5]. Irrigation and staking specifications belong inside that clock; they expire when the clock closes.

Watson 把定植當成吸收根重建進入周圍土壤的過程，以根伸長與樹冠水分狀態測量，而不是普遍日數 [4]。Hilbert 等發現，在許多城市研究中，年死亡率在首五年高於其後 [5]。灌溉與支撐規格屬於此時鐘之內；時鐘關閉它們即到期。

5. Limits

5. 限度

Large-calibre tropical field-grown stock, palms, and internally girdled systems have thinner published clocks than temperate container maples. Aftercare budgets that assume a two-year close-out in a dry or compacted city are hopes, not Watson's result.

大徑熱帶地栽苗、棕櫚與內環縊系統的已發表時鐘，薄於溫帶容器楓。在乾旱或壓實城市裡假設兩年收束的養護預算，是願望，不是 Watson 的結果。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. The non-negotiable planting landmark is:

1. 不可談判的種植地標是：

A. Flower colour.

A. 花色。

B. The root flare at finished grade.

B. 完成標高處的幹基膨大部。

C. t/R.

C. t/R。

D. A packet inoculant.

D. 袋裝接種劑。

2. Circling container roots should be:

2. 盤繞的容器根應當：

A. Planted intact as a feature.

A. 作為特色原封種下。

B. Corrected before backfill.

B. 在回填前矯正。

C. Painted.

C. 上漆。

D. Ignored if the taxon is popular.

D. 若分類群受歡迎就可忽略。

3. A deep narrow hole:

3. 又深又窄的穴：

A. Restores a forest profile.

A. 恢復森林剖面。

B. Recreates a container.

B. 再造一個容器。

C. Raises P50.

C. 提高 P50。

D. Sets 3-30-300.

D. 設定 3-30-300。

4. Watson's clock measures:

4. Watson 的時鐘測量：

A. Only invoice dates.

A. 只有發票日期。

B. Rebuilding of absorbing roots, not a universal day-count.

B. 吸收根重建，不是普遍日數。

C. Flash density.

C. 閃擊密度。

D. Wall 2.

D. 第 2 壁。

5. Hilbert et al. found mortality in years 0–5:

5. Hilbert 等發現第 0–5 年死亡率：

A. Lower than afterwards.

A. 低於其後。

B. Elevated relative to later years.

B. 相對往後年份升高。

C. Zero.

C. 為零。

D. Equal to P88.

D. 等於 P88。

6. Heavily amended backfill can:

6. 大幅改良的回填可以：

A. Only help.

A. 只會有幫助。

B. Recreate pot hydrology inside native soil.

B. 在原生土裡再造盆水文。

C. Replace the flare.

C. 取代膨大部。

D. Print new section.

D. 印出新截面。

7. Volcano mulch against the stem:

7. 抵住莖的火山狀覆蓋：

A. Is the A300 Part 2 specification.

A. 是 A300 第 2 部分規格。

B. Is the opposite of that specification.

B. 是該規格的反面。

C. Is a vulnerability curve.

C. 是脆弱性曲線。

D. Is Wall 4.

D. 是第 4 壁。

8. Aftercare specifications:

8. 養護規格：

A. Run forever by folklore.

A. 靠傳聞永遠運轉。

B. Belong inside the establishment clock and expire.

B. 屬於定植時鐘之內並到期。

C. Replace species selection.

C. 取代樹種選擇。

D. Set occupancy.

D. 設定佔用。

9. Burying the flare is:

9. 埋住膨大部是：

A. A matter of taste.

A. 品味問題。

B. A chronic stem-and-oxygen defect.

B. 慢性莖與氧缺陷。

C. Required in palms only.

C. 只在棕櫚中被要求。

D. A 10-20-30 rule.

D. 10-20-30 規則。

10. A planting target without an establishment survival budget:

10. 沒有定植存活預算的種植目標：

A. Is a complete canopy forecast.

A. 是完整樹冠預測。

B. Overstates future canopy.

B. 高估未來樹冠。

C. Equals Choat's safety margin.

C. 等於 Choat 安全邊際。

D. Is a tomogram.

D. 是斷層圖。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Gilman E.F. et al. container-root shaving and post-planting architecture.

[2] Watson G.W., Himelick E.B. Principles and Practice of Planting Trees and Shrubs. ISA.

[3] ANSI A300 Part 6 planting; Part 2 soils and mulch.

[4] Watson G.W. establishment as root reconnection rather than a calendar.

[5] Hilbert D.R. et al. (2019). Urban tree mortality. AUF 45: 167–200.

[6] Struve D.K. transplant shock and root:shoot rebalancing.

[7] Appleton B.L. circling-root and depth errors.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。